TCS NQT 2027 · SECTION 4 OF 5 · PART B · ADVANCED

Advanced Quant + Reasoning Complete Guide

Every topic explained from zero: rules, formulas, diagrams, solved examples and practice questions.

10	40+	100+	14
topics from basic to advanced	solved examples	practice questions	question mini test

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About this guide

Advanced Quantitative + Reasoning is the first section of Part B (Advanced) in the TCS NQT. It has 14 questions in 25 minutes (about 107 seconds each) at high difficulty, with no negative marking.

How each topic is organised

- **Start from zero:** the idea in plain words.
- **Formulas and rules:** a yellow box to revise quickly.
- **Diagrams:** pictures, charts and tables.
- **Solved examples:** step by step.
- **Shortcuts and mistakes** to avoid.
- **Practice questions:** Basic, Medium and Advanced. Answers are at the end.

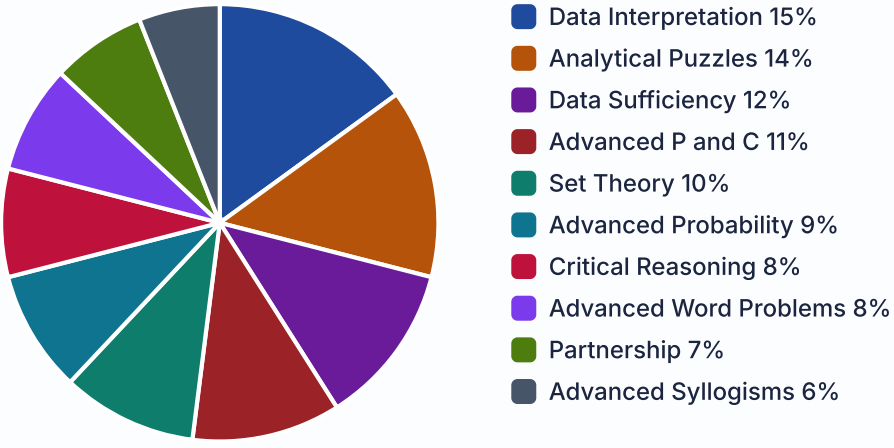
How to study

- Read one topic, then solve its practice set without looking at the answers.
- Check answers and note wrong ones in an error log.
- Revise the formula boxes daily.
- After all topics, take the mini test under the real time limit.
- Follow the priority: do **Must-do** topics first.

Topic priority and contents

1. Data Interpretation	Must-do	2. Set Theory and Venn Diagrams	Must-do
3. Data Sufficiency	Must-do	4. Analytical Puzzles	Must-do
5. Advanced Permutation and Combination	Must-do	6. Advanced Probability	Important
7. Partnership	Important	8. Critical Reasoning	Important
9. Advanced Syllogisms	Important	10. Advanced Word Problems	Important

Where to spend your practice time



Suggested share of practice time per topic. This is our preparation guidance, not an official TCS weightage.

Section strategy

Advanced Quantitative + Reasoning has 14 questions in 25 minutes (about 107 seconds each) at high difficulty. It is the first section of Part B (Advanced), and it counts for the Digital and Prime roles.

Time plan

Phase	Time	What to do
Scan	2 min	Look at all 14 and mark the easiest 8–9
Pass 1	14 min	Solve DI sets, puzzles and data sufficiency first
Pass 2	7 min	Return to harder ones
Final	2 min	Guess any left blank

Targets

- **Safe score:** 9–10 of 14 correct.
- **Good score:** 11 or more.
- No negative marking, so attempt every question.
- Choose by topic strength, not by order.

Speed habits

- Start with data sets: one chart can carry several questions.
- Approximate in DI instead of exact division.
- Test each statement alone in data sufficiency.
- Draw a table before reading puzzle clues twice.

Where marks are lost

- Reading a full DI set before the questions.
- Staying more than 3 minutes on one puzzle.
- Solving data sufficiency fully.
- Skipping the unit check.

01 ADVANCED QUANT + REASONING · MUST-DO

Data Interpretation

Data Interpretation (DI) asks questions from tables, bar charts, line graphs and pie charts. The skill is reading the data fast and using approximation instead of long calculations.

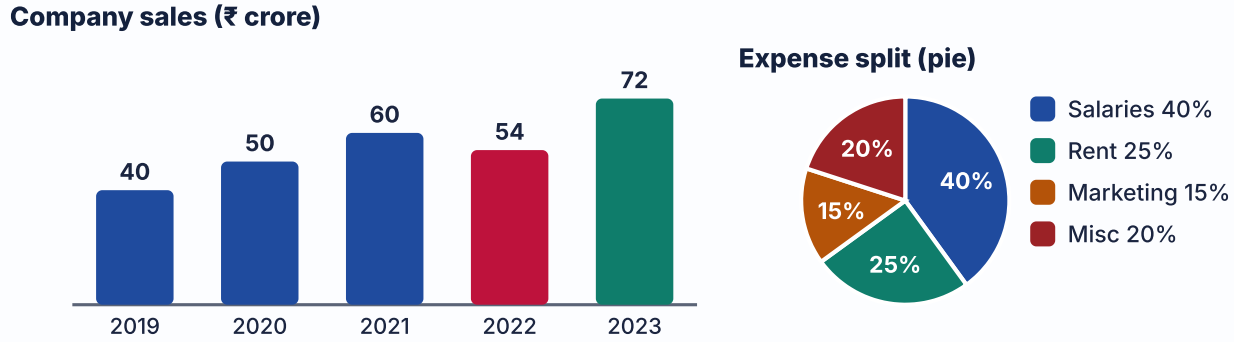
Start from zero

Read the chart first

Look at the title, the axis labels and the units before reading any question. Note whether the numbers are absolute or percentages, and whether they are in thousands or crores. Most mistakes come from misreading units.

Percentage change and ratio

Percentage change = (new − old)/old × 100. Ratio of two values = value1 : value2, simplified. Share = part/total × 100. Example from the bar chart: sales rose from 54 to 72, so the growth is (72 − 54)/54 × 100 = 33.33%.



A bar chart, a pie chart and the formulas used with them

Pie charts

A pie chart shows parts of a whole. 1% = 3.6°, so the angle of a sector = % × 3.6°. If total = ₹8,00,000 and Salaries is 40%, then Salaries = 0.40 × 8,00,000 = ₹3,20,000. If angles are given, % = angle/360 × 100.

Tables and caselets

In a table, find the row and column you need and avoid computing the rest. A caselet is a paragraph of data; convert it to a small table first. Average = sum/count. Percentage points are differences of percentages (from 20% to 25% = +5 points, but +25% growth).

Approximation tactics

Round numbers to one or two significant digits when options are far apart. Use fractions (25% = 1/4, 12.5% = 1/8) instead of decimals. Compare ratios by cross-multiplying instead of dividing.

Formula and rule box

- % change = (new − old)/old × 100
- Pie sector angle = % × 3.6°
- Share % = part/total × 100
- Average = sum/number of items
- Percentage points = difference of two percentages
- Ratio a : b = a/b
- Approximate: 48.7% of 623 ≈ 50% of 620

Step-by-step method

1. Read the title, axes and units.
2. Write the numbers you need in a small table.
3. Use fractions and rounding to estimate.
4. Check units (₹ lakh vs crore, thousands) at the end.

Solved examples

Example 1. Sales (₹ crore) were 40, 50, 60, 54, 72 in 2019–2023. Find the percentage growth from 2019 to 2020.

Solution: (50 − 40)/40 × 100 = 25%.

Answer: 25%

Example 2. Find the average sales for the five years.

Solution: Sum = 40 + 50 + 60 + 54 + 72 = 276. Average = 276/5 = 55.2 crore.

Answer: 55.2 crore

Example 3. Expenses: Salaries 40%, Rent 25%, Marketing 15%, Misc 20% of ₹8,00,000. Find the Salaries amount and the Marketing angle.

Solution: Salaries = 0.40 × 8,00,000 = ₹3,20,000. Marketing angle = 15 × 3.6° = 54°.

Answer: ₹3,20,000; 54°

Example 4. Which year had the highest percentage increase over the previous year?

Solution: 2020: 25%, 2021: 20%, 2022: −10%, 2023: 33.33%. The highest is 2023.

Answer: 2023

Shortcuts

- 25% = 1/4, 20% = 1/5, 12.5% = 1/8 for quick estimates.
- Compare ratios by cross-multiplication.
- Sector angle ÷ 3.6 gives the percentage.

Common mistakes

- Missing the unit (₹ lakh vs crore).
- Confusing percentage points with percent change.
- Calculating exactly when estimation is enough.

Practice questions: Data Interpretation

1. Marks in five subjects are 70, 80, 65, 90, 75. Find the average. **Basic**
2. A pie sector is 20%. Find its angle. **Basic**
3. Sales rose from 80 to 100. Percentage increase? **Basic**
4. A = 120, B = 150, C = 90. B is what percent more than C? **Basic**
5. Production was 200, 250 and 300 in 2020, 2021 and 2022. Ratio of 2020 to 2022? **Medium**
6. Pie sector angles are 90°, 120° and 150°. What percentage is the 150° sector? **Medium**
7. Revenue was 500, 600, then 540. Percentage change from year 2 to year 3? **Medium**
8. Sales were 40, 55, 66 in successive years. 66 is what % above 55? **Medium**
9. In 2022 a firm has 60 men and 40 women. In 2023 men rise 10% and women rise 25%. Percentage increase in total staff? **Advanced**
10. Company A sales: 120, 150, 180; Company B: 100, 120, 160. In which year does A exceed B by the highest percentage of B? **Advanced**

Answers: see the answer key at the end of this guide.

02 Set Theory and Venn Diagrams

ADVANCED QUANT + REASONING · MUST-DO

Set questions count how many people fall into overlapping groups. Draw the Venn diagram and fill it from the centre outwards.

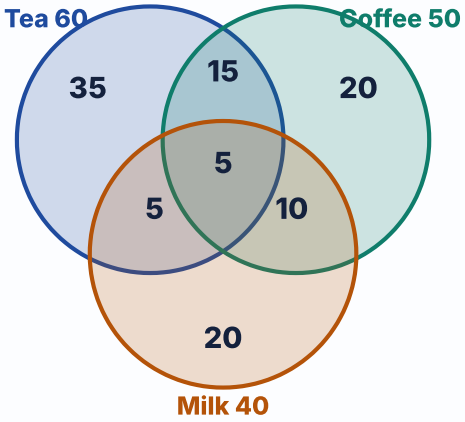
Start from zero

Two sets

n(A ∪ B) = n(A) + n(B) − n(A ∩ B). 'Neither' = total − n(A ∪ B). 'Only A' = n(A) − n(A ∩ B). Example: in a class of 60, 35 play cricket, 28 play football and 10 play both. Union = 35 + 28 − 10 = 53, so 7 play neither.

Three sets

n(A ∪ B ∪ C) = n(A) + n(B) + n(C) − n(A∩B) − n(B∩C) − n(A∩C) + n(A∩B∩C). Fill the diagram from the middle: first the triple overlap, then the pair-only regions (pair − triple), then the single-only regions.



Three-set Venn diagram, filled from the centre outwards

120 people. None = 10.

Start at the centre (all three = 5).

Pairs only: 20−5, 15−5, 10−5.

Singles: 60−(15+5+5) = 35

Exactly one = 35 + 20 + 20 = 75

Exactly and at least

Exactly two = sum of the three 'pair only' regions. **Exactly one** = sum of the 'only' regions. **At least one** = union. **None** = total − union.

Maximum and minimum overlap

With a universe of size U, the minimum overlap of A and B is n(A) + n(B) − U (if positive) and the maximum is the smaller set. Example: 70% like tea and 60% like coffee, so at least 70 + 60 − 100 = 30% like both.

Percentage versions

Treat percentages as numbers out of 100. Survey statements like '60% read paper X' translate directly into set sizes.

Formula and rule box

- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- $n(A \cup B \cup C) = \Sigma A - \Sigma \text{pairs} + \text{triple}$
- Only A = $n(A) - n(A \cap B) - n(A \cap C) + n(A \cap B \cap C)$
- Pair only (A and B) = $n(A \cap B) - n(A \cap B \cap C)$
- Neither = Total − $n(A \cup B \cup C)$
- Minimum overlap = $n(A) + n(B)$ − Total
- Maximum overlap = smaller set

Step-by-step method

1. Draw overlapping circles and a rectangle for the universe.
2. Fill the triple overlap first.
3. Compute pair-only regions, then single-only regions.
4. Check that the total equals the universe.

Solved examples

Example 1. A class has 60 students. 35 play cricket, 28 play football and 10 play both. How many play neither?

Solution: Union = 35 + 28 − 10 = 53. Neither = 60 − 53 = 7.

Answer: 7

Example 2. 120 people: Tea 60, Coffee 50, Milk 40; Tea∩Coffee 20, Coffee∩Milk 15, Tea∩Milk 10, all three 5. How many drink exactly one beverage?

Solution: Only tea = 60 − 20 − 10 + 5 = 35. Only coffee = 50 − 20 − 15 + 5 = 20. Only milk = 40 − 15 − 10 + 5 = 20. Total = 75.

Answer: 75

Example 3. From a universe of 50, set A has 40 members and B has 30. Find the least overlap.

Solution: 40 + 30 − 50 = 20.

Answer: 20

Example 4. 70% like tea and 60% like coffee. At least what percent like both?

Solution: 70 + 60 − 100 = 30%.

Answer: 30%

Shortcuts

- Always fill the centre first.
- Use the rectangle for the universe so you can see 'neither'.
- 'At least' and 'at most' overlap use the formula, not a diagram.

Common mistakes

- Subtracting the triple overlap too many times.
- Forgetting the 'neither' region.
- Treating 'only A' as n(A).

Practice questions: Set Theory and Venn Diagrams

1. $n(A) = 20$, $n(B) = 15$ and $n(A \cap B) = 5$. Find $n(A \cup B)$. **Basic**
2. In a class of 50, 30 take maths, 25 take science and 10 take both. How many take neither? **Basic**
3. In a group of 80, 50 speak Hindi and 40 speak English. If everyone speaks at least one, how many speak both? **Basic**
4. $n(A \cup B) = 70$, $n(A) = 45$ and $n(B) = 40$. Find $n(A \cap B)$. **Basic**
5. $n(A) = 45$ and $n(A \cap B) = 15$. Find the number of elements only in A. **Medium**
6. Of 100 students, 60 like pizza, 50 like burgers and 25 like both. How many like neither? **Medium**
7. $n(A) = 30$, $n(B) = 25$, $n(C) = 20$, $n(A \cap B) = 10$, $n(B \cap C) = 8$, $n(A \cap C) = 6$, $n(A \cap B \cap C) = 4$. Find $n(A \cup B \cup C)$. **Medium**
8. 90% like cricket and 80% like football. At least what percent like both? **Medium**
9. In a group of 100, 45 read English, 55 read Hindi and 10 read neither. How many read both? **Advanced**
10. A survey of 195 persons: 120 have a phone, 90 a laptop, 70 a tablet; phone∩laptop 40, laptop∩tablet 30, phone∩tablet 25, all three 10. How many have exactly two devices? **Advanced**

Answers: see the answer key at the end of this guide.

03 Data Sufficiency

ADVANCED QUANT + REASONING · MUST-DO

Data Sufficiency does not ask you to find the answer, only to decide whether the given statements contain enough information to find it.

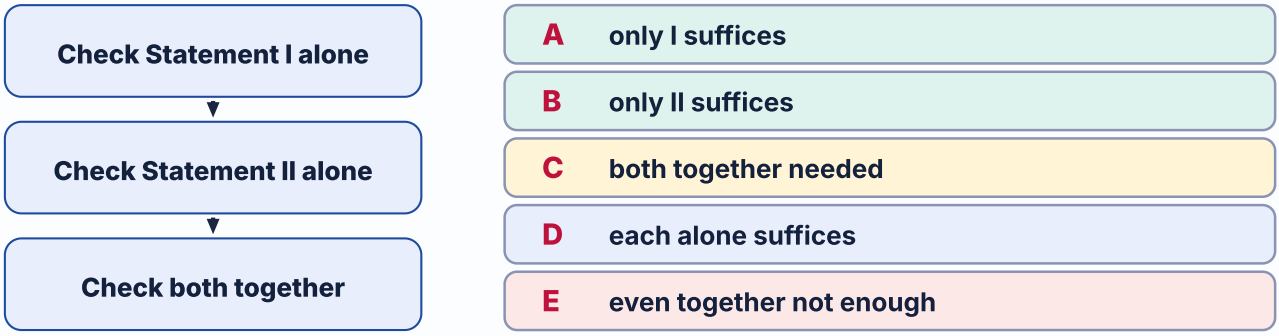
Start from zero

The format

A question is followed by two statements. You must decide whether the question can be answered (1) using Statement I alone, (2) using Statement II alone, (3) using both together, (4) using either alone, or (5) not even with both.

The five options

A: I alone is sufficient, II alone is not. **B:** II alone is sufficient, I alone is not. **C:** Both together are needed, neither alone is sufficient. **D:** Each statement alone is sufficient. **E:** Even together they are not sufficient.



Never solve the full problem: only decide whether the data would be enough to give ONE definite answer.

The decision path and the five answer options

What 'sufficient' means

The data must give exactly one definite answer. For a yes/no question, sufficient means always yes or always no. If the answer could be yes in one case and no in another, the data is insufficient.

The test routine

Test Statement I alone. Test Statement II alone. If neither is sufficient, combine them. Never carry information from I into the test of II. Do not compute the full answer; just see whether it is possible.

Typical traps

$x^2 = 16$ gives two values (±4). A multiple of 3 can be odd or even. 'x > 0' alone rarely fixes a number. When two statements are equivalent (say $2x = 8$ and $x = 4$), the answer is D.

Formula and rule box

- I alone → A; II alone → B
- Both needed → C
- Each alone → D
- Neither even together → E
- Sufficient means one unique answer
- Yes/No: always yes or always no

Step-by-step method

1. Rewrite what the question needs.
2. Test I alone, then II alone.
3. Combine only if needed.
4. Mark A, B, C, D or E.

Solved examples

Example 1. What is x? I. $x^2 = 16$. II. $x > 0$.

Solution: I gives x = 4 or −4. II alone gives no value. Together x = 4. So both are needed.

Answer: C

Example 2. Is n even? I. n/2 is an integer. II. n is a multiple of 3.

Solution: I says n is divisible by 2, so n is even: sufficient. II alone can be 3 (odd) or 6 (even).

Answer: A

Example 3. How old is Ravi? I. Ravi is 5 years older than Amit. II. Amit is 20.

Solution: I gives only a difference. II alone gives Amit's age. Together Ravi = 25.

Answer: C

Example 4. What is the speed of the train? I. It crosses a pole in 12 s. II. Its length is 180 m.

Solution: Speed = length/time. Neither alone is enough. Together 180/12 = 15 m/s.

Answer: C

Shortcuts

- Rewrite the question in the simplest form before testing statements.
- For yes/no, find one case giving yes and another giving no to prove insufficiency.
- Statements are always true: do not doubt them.

Common mistakes

- Combining I and II when one is already sufficient.
- Solving for the actual value and wasting time.
- Using outside information.

Practice questions: Data Sufficiency

1. What is y? I. y = 2x. II. x = 4. Basic

2. Is x positive? I. $x^2 = 9$. II. $x + 3 = 6$. Basic

3. What is the area of a rectangle? I. Perimeter is 24. II. Length is twice the width. Basic

4. Is n divisible by 6? I. n is divisible by 2. II. n is divisible by 3. Basic

5. What is the price of a pen? I. 3 pens cost ₹30. II. A pencil costs ₹5. Medium

6. How many students are in the class? I. 40% are girls. II. Boys number 36. Medium

7. Is $x > y$? I. $x = y + 2$. II. $y < 5$. Medium

8. What is the average of a, b, c? I. $a + b + c = 36$. II. $a = b = c$. Medium

9. A 200 m train crosses a platform. What is its speed? I. The platform is 100 m long. II. The crossing takes 15 s. Advanced

10. Is n a perfect square? I. n is 16 or 25. II. n is odd. Advanced

Answers: see the answer key at the end of this guide.

04 Analytical Puzzles

ADVANCED QUANT + REASONING · MUST-DO

Analytical puzzles are small logic problems with a unique answer. The plan is always the same: organise the facts in a table, place the definite clues first and eliminate.

Start from zero

Scheduling and ordering

Draw days or positions across the top and persons down the side. Place fixed facts first (B on Wednesday), then use relative clues (A the day before C) and eliminations. Exactly one arrangement should remain.

Truth-tellers and liars

Assume one person is truthful and follow the consequences. If you reach a contradiction, the assumption is wrong. A person who says 'we are both liars' cannot be truthful (that statement would make him a liar), so he is a liar and the other is truthful.

Five meetings A–E, one per day: B = Wed, E = Fri, D not Mon/Fri, A the day before C

Mon	Tue	Wed	Thu	Fri
A	C	B	D	E

Method: write the days, place fixed facts first (B, E), then test the remaining combinations.

Round robin of n teams → nC2 matches (6 teams: 15). Knockout of n teams → n − 1 matches.

A solved scheduling grid

Grouping and selection

List all possible groups, then cross out those that violate the rules. With 5 people choose 3 where P and Q cannot go together and R needs S: the valid teams are PRS, PST, QRS, QST and RST, so 5 teams.

Tournaments

In a round robin of n teams every pair plays once: nC2 matches (8 teams: 28). In a knockout of n teams: n − 1 matches. If every pair plays twice (home and away): n(n − 1) matches.

Number and logic puzzles

Translate the story into equations: three consecutive odd numbers with sum 75 give n − 2, n, n + 2 with 3n = 75, so the numbers are 23, 25, 27. A box with a wrong label must contain the other kind: a box labelled 'Mixed' with all wrong labels is pure.

Formula and rule box

- Round robin = nC2
- Knockout = n − 1
- Home and away = n(n − 1)
- Three consecutive numbers: n − 1, n, n + 1
- Elimination: place definite clues first

Step-by-step method

1. Draw a table or list the objects and positions.
2. Place the definite clues.
3. Use relative and negative clues to eliminate.
4. Verify that every clue is satisfied.

Solved examples

Example 1. Five meetings A–E are held Monday to Friday, one a day. B is on Wednesday, E on Friday, D is not on Monday or Friday and A is the day before C. On which day is D?

Solution: Free days: Mon, Tue, Thu for A, C, D. A is the day before C, so A = Mon and C = Tue. D = Thu.

Answer: Thursday

Example 2. A says 'We are both liars.' Which is true?

Solution: If A were truthful, the statement would be false. So A is a liar, and B must be truthful.

Answer: A liar, B truthful

Example 3. 5 people P–T: choose 3 where P and Q cannot go together, and R can go only if S goes. How many teams?

Solution: Valid: PRS, PST, QRS, QST, RST = 5.

Answer: 5

Example 4. In a round-robin tournament of 8 teams, how many matches are played?

Solution: 8C2 = 28.

Answer: 28

Shortcuts

- Always put the fixed facts in the table first.
- For liars, assume a statement is true and look for a contradiction.
- Use formulas for tournaments: nC2 or n − 1.

Common mistakes

- Starting with a relative clue.
- Forgetting to check all clues at the end.
- Mixing round-robin with knockout formulas.

Practice questions: Analytical Puzzles

1. A round-robin tournament has 6 teams. How many matches? Basic
2. A says 'B is truthful.' B says 'A and I are of opposite types.' What are they? Basic
3. P's birthday is Tuesday, Q's is the day after P's and R's is the day before P's. Four friends have birthdays on Monday to Thursday. On which day is S's? Basic
4. From A, B, C, D choose 2 where A and B cannot go together. How many ways? Basic
5. Five books in a pile: red above blue, green between red and blue, yellow at the bottom, black at the top. Which is in the middle? Medium

6. Three boxes labelled Apples, Oranges and Mixed all carry wrong labels. A fruit from the 'Mixed' box is an apple. What does it hold? Medium

7. Amit is taller than Bina but shorter than Chitra. Dev is taller than Chitra. Who is the shortest? Medium

8. Three consecutive odd numbers sum to 75. Find the largest. Medium

9. A knockout tournament has 16 teams. How many matches? Advanced

10. Five teams play each other twice (home and away). How many matches? Advanced

Answers: see the answer key at the end of this guide.

05 ADVANCED QUANT + REASONING · MUST-DO

Advanced Permutation and Combination

Advanced P and C questions add restrictions: people who must sit together, digits that cannot repeat, groups with at least one member of a kind, and objects that are identical.

Start from zero

Arrangements with repeated letters

If n letters have repeats p, q, r, the arrangements = n!/(p! q! r!). Example: LEADER has 6 letters with E twice: 6!/2! = 360. If the vowels must stay together, glue E, A, E into a block: arrangements inside the block = 3!/2! = 3, and the 4 units (block, L, D, R) give 4! = 24. Total = 72.

The glue method

If some persons must be together, treat them as one unit, arrange the units, then arrange the persons inside the unit. 5 boys and 3 girls in a row with the girls together: 6! × 3! = 4,320.

Stars and bars: 10 identical chocolates, 3 children → place 2 dividers



Total positions = 12, choose 2 for the bars → 12C2 = 66

Glue method: treat people who must be together as ONE unit,
arrange the units, then arrange the persons inside the unit.
5 boys + 3 girls, girls together: 6! × 3! = 4320

Stars and bars, and the glue method

Number formation

For an r-digit number from digits without repetition, fill the restricted places first (first digit not zero, last digit even). Example: 4-digit numbers from 0–9 without repeats: 9 × 9 × 8 × 7 = 4,536.

Selection with conditions

'At least' problems use complement: total − (fewer than needed). A committee of 4 from 6 men and 4 women with at least 2 women: C(10,4) − C(6,4) − 4 × C(6,3) = 210 − 15 − 80 = 115.

Distribution of identical objects

To share n identical objects among r people (each can get zero): C(n + r − 1, r − 1). With at least one each: C(n − 1, r − 1). 10 chocolates among 3 children: C(12, 2) = 66; at least one each: C(9, 2) = 36.

Circular arrangements

n people around a table: (n − 1)!. If two particular people must be together: treat them as one unit, so (n − 2)! × 2 for n people. Six people with A and B together: 4! × 2 = 48.

Formula and rule box

- Repeats: n!/(p! q! ...)
- Circular: (n − 1)!
- Glue: (units)! × (arrangements inside)
- At least: total − complement
- Identical objects (zero allowed): C(n + r − 1, r − 1)
- Identical objects (≥ 1 each): C(n − 1, r − 1)
- Diagonals of an n-gon: nC2 − n

Step-by-step method

1. Identify arrangement or selection.
2. Handle restrictions first (glue, first digit).
3. Use complement for 'at least'.
4. Check small cases to confirm the method.

Solved examples

Example 1. How many arrangements of LEADER have the vowels together?

Solution: Vowels E, A, E block: 3!/2! = 3. Units: block, L, D, R → 4! = 24. Total 72.

Answer: 72

Example 2. 5 boys and 3 girls stand in a row with the girls together. How many ways?

Solution: Block of girls: 6! × 3! = 720 × 6 = 4,320.

Answer: 4,320

Example 3. How many 4-digit numbers have no repeated digit?

Solution: First digit 9 ways (not 0), then 9, 8, 7: 9 × 9 × 8 × 7 = 4,536.

Answer: 4,536

Example 4. Distribute 10 identical chocolates among 3 children (zero allowed).

Solution: C(12, 2) = 66.

Answer: 66

Shortcuts

- Fix restricted positions first.
- Use the complement for 'at least one'.
- For identical objects use stars and bars.

Common mistakes

- Forgetting to arrange inside the glued group.
- Allowing zero as the first digit.
- Using nPr when order does not matter.

Practice questions: Advanced Permutation and Combination

1. In how many ways can 7 people stand in a row? Basic

2. How many arrangements does BANANA have? Basic
3. How many even 3-digit numbers can be formed from 1–5 without repetition? Basic
4. A committee of 4 is formed from 6 men and 4 women with at least 2 women. How many ways? Basic
5. How many arrangements does PAPER have? Medium
6. How many diagonals does a decagon have? Medium
7. Select 3 people from 5 boys and 4 girls with at least one girl. How many ways? Medium
8. In how many ways can 6 people be divided into 3 unlabelled pairs? Medium
9. How many 4-letter words can be formed from LOGIC without repetition? Advanced
10. Distribute 12 identical balls into 4 boxes with at least one in each. Advanced

Answers: see the answer key at the end of this guide.

06 Advanced Probability

ADVANCED QUANT + REASONING · IMPORTANT

Advanced probability adds conditions: draws without replacement, choosing among bags, and 'given that' questions. Trees make these simple.

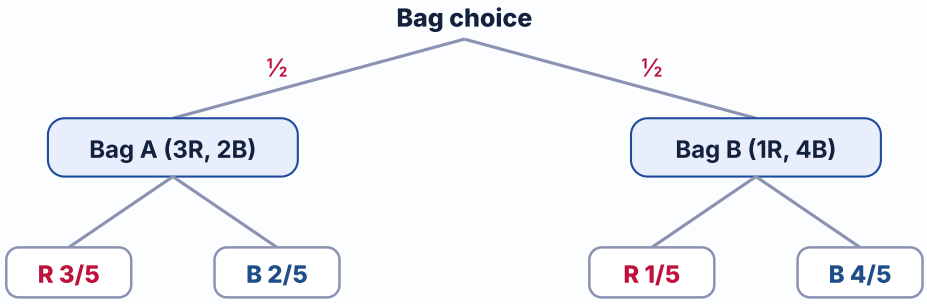
Start from zero

Conditional probability

P(A | B) = P(A and B)/P(B): the probability of A when you already know B has happened. Example: a die shows an even number; the chance it is greater than 3 is 2/3 (outcomes 4, 6 out of 2, 4, 6).

Without replacement

The probability changes after each draw. Two hearts from a deck: (13/52) × (12/51) = 1/17. Use combinations for multi-object draws: P(all red) = C(5,3)/C(9,3).



P(red) = ½ × 3/5 + ½ × 1/5 = 2/5

Given red, P(from A) = (½ × 3/5)/(2/5) = 3/4

A probability tree for two bags

Tree diagrams and total probability

If a bag is chosen at random, then a ball drawn, multiply along each branch and add the branches that give the wanted outcome. Using the tree: P(red) = ½ × 3/5 + ½ × 1/5 = 2/5.

Bayes in plain words

Given that a red ball was drawn, the chance it came from bag A is (the branch via A)/(all red branches) = (½ × 3/5)/(2/5) = 3/4.

Expectation

Expected value = Σ (value × probability). A die: (1 + 2 + … + 6)/6 = 3.5. A game paying ₹10 on heads and −₹4 on tails: ½ × 10 − ½ × 4 = ₹3.

Formula and rule box

- P(A | B) = P(A ∩ B)/P(B)
- P(A and B) = P(A) × P(B | A)
- Total probability = Σ (branch products)
- Bayes: P(A | R) = P(A) P(R | A)/P(R)
- Expected value = Σ x · P(x)
- P(at least one) = 1 − P(none)

Step-by-step method

1. Decide whether draws are with or without replacement.
2. Draw a tree for multi-stage problems.
3. Multiply along branches and add parallel branches.
4. For conditional questions, restrict the sample space.

Solved examples

Example 1. Two cards are drawn without replacement. Find P(both are hearts).

Solution: (13/52) × (12/51) = 156/2652 = 1/17.

Answer: 1/17

Example 2. A die is rolled. Given that the number is even, find P(number > 3).

Solution: Even: 2, 4, 6. Greater than 3: 4, 6. So 2/3.

Answer: 2/3

Example 3. Bag A has 3 red and 2 blue, bag B has 1 red and 4 blue. A bag is chosen at random and a ball drawn. Find P(red).

Solution: ½ × 3/5 + ½ × 1/5 = 2/5.

Answer: 2/5

Example 4. Given the ball is red, find the probability it came from bag A.

Solution: (½ × 3/5)/(2/5) = 3/4.

Answer: 3/4

Shortcuts

- Use the complement for 'at least one'.
- For 'given that', shrink the sample space to the given event.
- Combinations are quicker than multiplying fractions for several draws.

Common mistakes

- Using the same probability after a draw without replacement.
- Forgetting to divide by P(B) in conditional problems.
- Adding probabilities for 'and' events.

Practice questions: Advanced Probability

1. Two coins are tossed. Given at least one head, find P(both heads). Basic
2. Two dice are rolled and the first shows 3. Find P(sum = 8). Basic
3. A bag has 4 red and 6 blue balls. Two are drawn without replacement. Find P(both blue). Basic
4. A coin is tossed 4 times. Find P(at least one head). Basic
5. A card is drawn from a deck. Find P(red or a face card). Medium
6. Three balls are drawn from 5 red and 4 blue. Find P(all red). Medium
7. A number is chosen from 1 to 20. Find P(divisible by 3 or 5). Medium

8. Two dice are rolled. Find P(product is even). Medium

9. A box has 3 defective items in 10. Two are drawn without replacement. Find P(exactly one defective). Advanced

10. A game pays ₹10 for heads and −₹4 for tails. Find the expected value per toss. Advanced

Answers: see the answer key at the end of this guide.

07 ADVANCED QUANT + REASONING · IMPORTANT

Partnership

In a partnership, profit is shared in proportion to capital and time. Everything reduces to multiplying money by months.

Start from zero

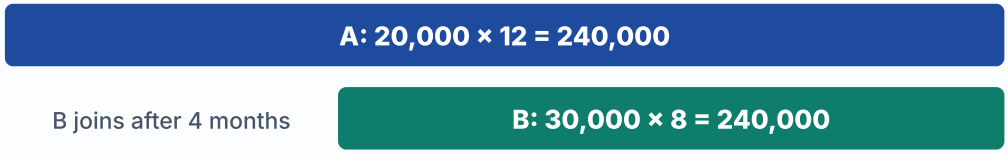
The basic rule

Profit ratio = (capital$_1$ × time$_1$) : (capital$_2$ × time$_2$). If all capitals stay for the same time, the ratio is just the ratio of capitals. If all capitals are equal, it is the ratio of time.

Partners who join or leave

Treat each period with its own capital. A invests ₹20,000 for 12 months, B invests ₹30,000 and joins after 4 months (so stays 8 months): 240,000 : 240,000 = 1 : 1.

Profit share = capital × time. A: 20,000 for 12 months · B: 30,000 for 8 months



Ratio A : B = 1 : 1. Always multiply capital by the months it stays in the business.

Investment × time decides the share

Changing capital

If A doubles his capital after 6 months and B keeps a steady capital, add the products for each period. A: 3 × 6 + 6 × 6 = 54; B: 5 × 12 = 60; the ratio is 9 : 10.

Working and sleeping partners

A working partner may get a salary or a percentage of profit first. The rest is divided by capital × time. A (working) gets 10% of ₹10,000 = ₹1,000 as salary; the remaining ₹9,000 is shared 3 : 2, so A gets 6,400 in total and B gets 3,600.

Fractional capitals

If capitals are fractions (1/2, 1/3, 1/6), convert to a common whole (3, 2, 1 parts).

Formula and rule box

- Profit ratio = Σ (capital × time)
- Equal time → ratio of capitals
- Equal capital → ratio of times
- Salary or commission comes first, rest by ratio
- Months of stay = 12 − months before joining

Step-by-step method

1. List each partner's capital and time.
2. Multiply capital × time for each period and add.
3. Simplify the ratio.

4. Apply it to the profit (after salaries).

Solved examples

Example 1. A invests ₹5,000 for 12 months and B invests ₹8,000 for 6 months. Profit is ₹18,000. Find A's share.

Solution: A: 60,000, B: 48,000, ratio 5 : 4. A = 18,000 × 5/9 = ₹10,000.

Answer: ₹10,000

Example 2. A invests ₹20,000. After 4 months B joins with ₹30,000. Find the profit ratio at year end.

Solution: A: 20,000 × 12 = 240,000. B: 30,000 × 8 = 240,000. Ratio 1 : 1.

Answer: 1 : 1

Example 3. A (working partner) gets 10% of the profit of ₹10,000 as salary. The rest is shared 3 : 2. Find A's total.

Solution: Salary 1,000. Remaining 9,000: A gets 5,400. Total 6,400.

Answer: ₹6,400

Example 4. A and B invest in the ratio 3 : 5. After 6 months A doubles his capital. Find the profit ratio for the year.

Solution: A: 3 × 6 + 6 × 6 = 54. B: 5 × 12 = 60. Ratio 54 : 60 = 9 : 10.

Answer: 9 : 10

Shortcuts

- Multiply capital by months, not by years, if the time is in months.
- If someone joins after x months, he stays 12 − x months.
- Remove common factors before dividing the profit.

Common mistakes

- Ignoring the time for which the capital stays.
- Subtracting the salary after dividing by ratio.
- Using years for one partner and months for another.

Practice questions: Partnership

1. A invests ₹6,000 and B ₹4,000 for 12 months each. Profit ₹5,000. Find A's share. Basic
2. A invests ₹4,000 for 6 months and B ₹6,000 for 4 months. Find the ratio. Basic
3. Capitals are in the ratio 2 : 3 for the same time. Profit ₹7,500. Find B's share. Basic
4. A invests ₹8,000 for 12 months. B invests ₹12,000 for 6 months. Profit ₹14,000. Find A's share. Basic
5. Three partners invest 1/2, 1/3 and 1/6 of the capital. Profit ₹6,000. Find the shares. Medium
6. A invests ₹10,000 for 8 months and B ₹8,000 for 10 months. Find the ratio. Medium
7. A, B, C invest ₹2,000, ₹3,000, ₹5,000. Profit ₹12,000. Find C's share. Medium
8. Profit ₹20,000. A (working) gets 10% first; the rest is shared 1 : 2 (A : B). Find B's share. Medium

9. A invests ₹x for 12 months and B invests ₹(x + 2000) for 6 months. They share profit equally. Find x. **Advanced**

10. A and B invest ₹6,000 and ₹9,000. After 4 months A adds ₹3,000 and B withdraws ₹3,000. Year-end profit ₹24,000. Find A's share. **Advanced**

Answers: see the answer key at the end of this guide.

08 Critical Reasoning

ADVANCED QUANT + REASONING · IMPORTANT

Critical reasoning questions give a short argument and ask you to find an assumption, strengthen or weaken it, draw an inference or choose a course of action.

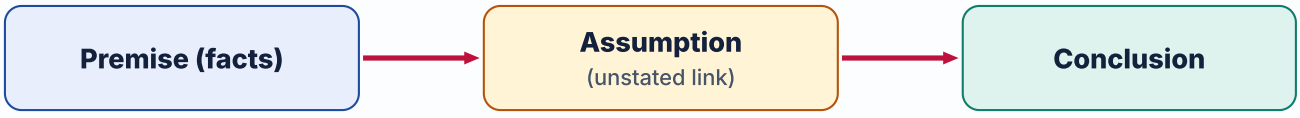
Start from zero

Parts of an argument

Premise: the facts given. **Conclusion:** what the author wants you to believe. **Assumption:** an unstated link that must be true for the premise to support the conclusion.

Assumption

Ask: what must be true for this argument to work? 'Buy Brand X because dentists recommend it' assumes that dentists' recommendations are reliable.



Strengthen: gives support to the link or shows it works elsewhere. Weaken: shows another cause or breaks the li

Inference: must be true from the facts alone. Course of action: a practical step that solves the stated problem.

Premise, assumption and conclusion

Strengthen and weaken

To strengthen, support the link (a similar case worked, a test confirmed it, alternative causes are ruled out). To weaken, show an alternative cause, a flaw or an exception. 'Rina lost weight after the pill, so the pill works' is weakened by 'she also started going to the gym'.

Inference and conclusion

An inference must be true from the facts alone. 'All employees who completed the training got promoted. Meena completed the training.' → Meena was promoted. A conclusion is a statement the facts best support.

Course of action

Pick a practical step that directly addresses the problem and does not create new problems. 'Many children miss school for lack of transport' → provide school buses (not 'close the school').

Cause and effect, paradox

Look for the most direct cause (heavy rain → rise in umbrella sales). For a paradox ('profit rose though sales fell'), choose the option that explains both facts (costs fell more, or prices rose).

Formula and rule box

- Premise → (assumption) → conclusion
- Strengthen: supports the link
- Weaken: gives another cause or breaks the link
- Inference: must be true from the facts
- Course of action: practical and direct
- Paradox: explains both facts

Step-by-step method

1. Find the conclusion and the premise.
2. Spot the gap between them.
3. Test each option against the gap.
4. Choose the option that fits the question type.

Solved examples

Example 1. 'Our company should adopt a four-day week because it will boost morale.' Find an assumption.

Solution: The argument assumes that a shorter week raises morale and that higher morale is desirable.

Answer: Morale rises with a four-day week

Example 2. 'City X should build a flyover to reduce jams.' Which statement strengthens it?

Solution: A similar flyover in City Y reduced jams by 40%.

Answer: The City Y result

Example 3. 'Rina lost 5 kg after taking the pill, so the pill works.' Which statement weakens it?

Solution: She also changed her diet and joined a gym at the same time.

Answer: Another cause

Example 4. 'All employees who completed the training were promoted. Meena completed the training.' What follows?

Solution: If completing training leads to promotion, Meena was promoted.

Answer: Meena was promoted

Shortcuts

- Identify the question type first.
- Look for the unstated link between premise and conclusion.
- Eliminate options that are extreme or irrelevant.

Common mistakes

- Choosing a statement that is merely true but irrelevant.
- Confusing strengthen with weaken.
- Using outside knowledge instead of the passage.

Practice questions: Critical Reasoning

1. Many children are absent from school for lack of transport. Course of action: I. Provide school buses. II. Close the school. Which follows? **Basic**
2. Umbrella sales rose sharply in July. What is the most likely cause? **Basic**
3. 'Students who sleep 8 hours score higher, so sleep causes higher scores.' What weakens it? **Basic**
4. 'Plants near music grow faster, so music helps growth.' What strengthens it? **Basic**
5. 'Buy Brand X toothpaste as dentists recommend it.' Find the assumption. **Medium**
6. Rainfall fell 20% this year and reservoirs are at 40%. Which conclusion is best supported? **Medium**
7. A company's profit rose though sales fell. What explains this? **Medium**
8. 'All A-grade students get scholarships. Sita did not get one.' What follows? **Medium**
9. A factory's smoke is making residents ill. Which action is best? **Advanced**
10. 'Crime fell after CCTV installation, so CCTV reduces crime.' What weakens it? **Advanced**

Answers: see the answer key at the end of this guide.

09

ADVANCED QUANT + REASONING · IMPORTANT

Advanced Syllogisms

Advanced syllogisms have three or four statements and use possibility and either-or conclusions. The Venn method from the basic topic still works; you simply draw more circles.

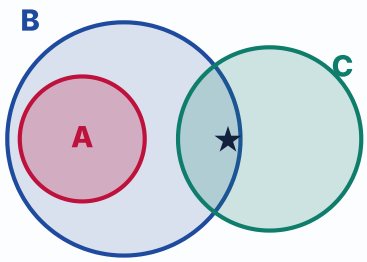
Start from zero

Chains of statements

Combine statements one pair at a time. All A are B and All B are C gives All A are C. Then combine the result with the next statement. Some A are B and All B are C gives Some A are C. Two 'Some' statements give nothing definite.

Possibility conclusions

A 'possibility' conclusion is true if it could happen in at least one valid diagram and is not contradicted by the statements. Some A are B, All B are C: the possibility 'All A are C' is true because A could sit entirely inside C.



All A are B · Some B are C (★ = overlap)

Some A are C? NOT certain (A may sit away from C)

Some C are B ✓ (conversion) · Some B are C ✓

Test every conclusion by trying to move the circles.

Venn diagram of a three-statement syllogism

Either-or

If two conclusions with the same subject and predicate form a complementary pair, such as 'All A are B' and 'Some A are not B', exactly one must be true. Then 'Either I or II follows'. Likewise 'Some A are C' and 'No A is C' for two 'some' statements.

Reverse thinking

If a conclusion needs two items to share a circle, check every statement for a 'No' that separates them. A single 'No' in the chain kills definite positive conclusions.

Venn tips

Draw the most restrictive statement first (All and No), then add Some by overlapping. Test each conclusion by moving circles; if it fails in even one valid diagram, it does not follow.

Formula and rule box

- All + All → All
- All + Some → no (use Some + All)
- Some + All → Some
- All + No → No
- No + All → Some not
- Some + Some → none
- Complementary pairs → either-or

Step-by-step method

1. Draw the circles for the first pair.
2. Add the next statement without breaking the first.
3. Test each conclusion in every valid diagram.
4. Decide: follows, does not follow, possible, or either-or.

Solved examples

Example 1. Statements: All A are B; Some B are C; Some C are D. Conclusions: I. Some A are C II. Some B are D III. Some D are B.

Solution: Two 'some' links give no definite result, and A need not touch C. None follows.

Answer: None follows

Example 2. Statements: Some A are B; All B are C. Possibility: All A are C?

Solution: A could lie entirely inside C, so it is possible.

Answer: Possible

Example 3. Statements: All A are B; All B are C; Some C are D; No D is E. Conclusions: I. All A are C II. Some C are not E III. No A is E.

Solution: I follows by the chain. Some C are D and no D is E, so II follows. III is not definite.

Answer: I and II follow

Example 4. Statement: Some A are B. Conclusions: I. All A are B II. Some A are not B.

Solution: They are contradictory, and one of them must be true. So either I or II.

Answer: Either I or II

Shortcuts

- Use letter chains: write the statements as A–B–C–D on rough paper.
- One 'No' in the middle blocks all definite positive conclusions across it.
- Possibility needs only one valid diagram.

Common mistakes

- Reversing 'All' into 'All'.
- Treating two 'Some' statements as a link.
- Forgetting the either-or case.

Practice questions: Advanced Syllogisms

1. All A are B; All B are C. I. Some C are A. II. Some A are not C. Basic
2. Some A are B; Some B are C; All C are D. I. Some A are D. II. Some B are D. Basic
3. No A is B; All B are C. I. Some C are not A. II. No C is A. Basic
4. All A are B; Some B are C; No C is D. I. Some B are not D. II. No A is D. Basic
5. Some A are B. Possibilities: I. All A are B. II. No A is B. Medium
6. All A are B; All C are B. I. All A are C. II. There is a possibility that some A are C. Medium
7. Some A are B; All B are C; Some C are D. I. Some A are C. II. Some A are D. III. Some B are D. Medium
8. All A are B; No B is C; All C are D. I. No A is C. II. Some D are not A. III. No A is D. Medium
9. Some A are B; No B is C. I. Some A are not C. II. No A is C. Advanced
10. Some A are B; Some B are C. I. Some A are C. II. No A is C. Advanced

Answers: see the answer key at the end of this guide.

10 Advanced Word Problems

ADVANCED QUANT + REASONING · IMPORTANT

These multi-step problems combine ideas from the basic Numerical topics: series, work with leaves, remainders and relative speed. They reward a calm, step-by-step setup.

Start from zero

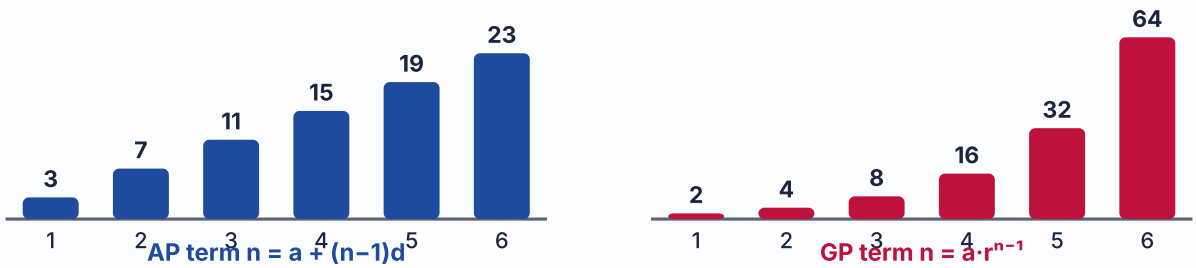
Arithmetic and geometric progressions

AP: each term adds a fixed d. Term n = a + (n − 1)d. Sum of n terms = n/2 [2a + (n − 1)d]. GP: each term multiplies by r. Term n = a r^{n-1}. Sum = a(r^n − 1)/(r − 1) for r > 1. Sum of the first n natural numbers = n(n + 1)/2 and of the first n odd numbers = n^2.

Work with early leaving

If someone leaves before completion, let total time = T. A works (T − 3) days, B works T days. A 12 days, B 18 days, A leaves 3 days early: (T − 3)/12 + T/18 = 1, which gives T = 9.

AP: 3, 7, 11, … adds 4 each time. GP: 2, 4, 8, … doubles each time.



Sum AP = n/2 [2a + (n−1)d] Sum GP = a(r^n − 1)/(r − 1)

AP grows by addition and GP by multiplication

Pipes closed midway

Split the time into stages: in stage 1 both pipes work; in stage 2 only one pipe works. Add the fractions of the tank filled in each stage.

Remainder patterns

Use cycles or negative remainders: 7 ≡ 1 (mod 6), so 7^{100} leaves 1; 5 ≡ −1 (mod 6), so 5^{100} leaves 1. 3^4 = 81 ≡ 1 (mod 5), so 3^{20} leaves 1.

Relative speed with a runner

Two moving objects in opposite directions add speeds. A 150 m train passes a man running opposite at 6 km/h in 6 s: relative speed = 150/6 = 25 m/s = 90 km/h, so the train's speed = 90 − 6 = 84 km/h.

Mixtures in stages

Repeated replacement reduces the original ingredient by the same fraction each time. 30 L of milk and water 2 : 1: after removing 10 L and adding water, milk left = 20 × (1 − 10/30) = 13.33 L.

Formula and rule box

- AP term: $a + (n - 1)d$; sum $n/2\ [2a + (n - 1)d]$
- GP term: $a\ r^{n-1}$; sum $a(r^n - 1)/(r - 1)$
- $1 + 2 + \ldots + n = n(n + 1)/2$; $1 + 3 + \ldots + (2n - 1) = n^2$
- Work with leaving: Σ (days worked/days alone) = 1
- Remainder: use cycles or $a \equiv \pm 1$
- Opposite directions: add speeds

Step-by-step method

1. Identify the underlying basic topic.
2. Write the unknown as T or x and set an equation.
3. Solve step by step.
4. Check by substitution.

Solved examples

Example 1. Find the sum of the first 20 terms of 3, 7, 11, ...

Solution: $a = 3$, $d = 4$. Sum $= 20/2\ [6 + 19 \times 4] = 10 \times 82 = 820$.

Answer: 820

Example 2. Find the sum of 2 + 4 + 8 + ... (6 terms).

Solution: $a = 2$, $r = 2$. Sum $= 2(2^6 - 1) = 126$.

Answer: 126

Example 3. A can do a job in 12 days, B in 18 days. They start together but A leaves 3 days before the work is finished. Find the total time.

Solution: $(T - 3)/12 + T/18 = 1 \rightarrow 3(T - 3) + 2T = 36 \rightarrow T = 9$.

Answer: 9 days

Example 4. A 150 m train passes a man running opposite at 6 km/h in 6 s. Find the train's speed.

Solution: Relative speed $= 150/6 = 25$ m/s $= 90$ km/h. Train $= 90 - 6 = 84$ km/h.

Answer: 84 km/h

Shortcuts

- Recognise AP versus GP from the first two differences.
- Use ±1 remainders for easy power problems.
- Convert units once at the start.

Common mistakes

- Using the GP formula for $r < 1$ without adjusting.
- Counting T days for both workers when one left early.
- Forgetting to subtract the man's speed.

Practice questions: Advanced Word Problems

1. Find the sum of the first 10 natural numbers. Basic

2. Find the 20th term of 5, 9, 13, ... Basic

3. Find the sum of the 5-term GP 3, 6, 12, ... Basic

4. A pipe fills a tank in 8 h and another empties it in 12 h. Both open for 2 h, then the outlet is closed. Time for the inlet to finish filling? Basic

5. A can do a job in 15 days and B in 10 days. They work together for 4 days. How many days does A need to finish the rest alone? Medium

6. Find the remainder of 7^{100} divided by 6. Medium

7. A boat goes 15 km downstream in 3 h and 9 km upstream in 3 h. Find the stream speed. Medium

8. A 30 L mixture of milk and water is 2 : 1. 10 L is removed and replaced with water. Find the milk left. Medium

9. Find 1 + 3 + 5 + ... (15 terms). Advanced

10. Two trains of 200 m and 150 m run in the same direction at 72 and 54 km/h. Find the time for the faster to pass the slower. Advanced

Answers: see the answer key at the end of this guide.

Cheat sheet

Every key rule and formula of the section on a few pages. Revise it daily in the last two weeks.

1. Data Interpretation

- % change = (new − old)/old × 100
- Pie sector angle = % × 3.6°
- Share % = part/total × 100
- Average = sum/number of items
- Percentage points = difference of two percentages
- Ratio a : b = a/b
- Approximate: 48.7% of 623 ≈ 50% of 620

2. Set Theory and Venn Diagrams

- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- $n(A \cup B \cup C) = \Sigma A - \Sigma$pairs + triple
- Only A = $n(A) - n(A \cap B) - n(A \cap C) + n(A \cap B \cap C)$
- Pair only (A and B) = $n(A \cap B) - n(A \cap B \cap C)$
- Neither = Total − $n(A \cup B \cup C)$
- Minimum overlap = n(A) + n(B) − Total
- Maximum overlap = smaller set

3. Data Sufficiency

- I alone → A; II alone → B
- Both needed → C
- Each alone → D
- Neither even together → E
- Sufficient means one unique answer
- Yes/No: always yes or always no

4. Analytical Puzzles

- Round robin = nC2
- Knockout = n − 1
- Home and away = n(n − 1)
- Three consecutive numbers: n − 1, n, n + 1
- Elimination: place definite clues first

5. Advanced Permutation and Combination

- Repeats: n!/(p! q! ...)
- Circular: (n − 1)!
- Glue: (units)! × (arrangements inside)
- At least: total − complement
- Identical objects (zero allowed): C(n + r − 1, r − 1)
- Identical objects (≥ 1 each): C(n − 1, r − 1)
- Diagonals of an n-gon: nC2 − n

6. Advanced Probability

- $P(A \mid B) = P(A \cap B)/P(B)$
- $P(A \text{ and } B) = P(A) \times P(B \mid A)$
- Total probability = Σ (branch products)
- Bayes: $P(A \mid R) = P(A)\,P(R \mid A)/P(R)$
- Expected value = $\Sigma\, x \cdot P(x)$
- P(at least one) = 1 − P(none)

7. Partnership

- Profit ratio = Σ (capital × time)
- Equal time → ratio of capitals
- Equal capital → ratio of times
- Salary or commission comes first, rest by ratio
- Months of stay = 12 − months before joining

8. Critical Reasoning

- Premise → (assumption) → conclusion
- Strengthen: supports the link
- Weaken: gives another cause or breaks the link
- Inference: must be true from the facts
- Course of action: practical and direct
- Paradox: explains both facts

9. Advanced Syllogisms

- All + All → All
- All + Some → no (use Some + All)
- Some + All → Some
- All + No → No
- No + All → Some not
- Some + Some → none
- Complementary pairs → either-or

10. Advanced Word Problems

- AP term: a + (n − 1)d; sum n/2 [2a + (n − 1)d]
- GP term: $a r^{n-1}$; sum $a(r^n - 1)/(r - 1)$
- $1 + 2 + \ldots + n = n(n + 1)/2$; $1 + 3 + \ldots + (2n - 1) = n^2$
- Work with leaving: Σ (days worked/days alone) = 1
- Remainder: use cycles or $a \equiv \pm 1$
- Opposite directions: add speeds

Mini test: 14 questions in 25 minutes

14 questions in 25 minutes. Target: 10 or more correct. No negative marking.

1. Sales were 54 and then 72. Find the percentage growth.
2. A pie sector represents 15%. Find its angle.
3. In a group of 50, 30 like A and 28 like B, and all like at least one. How many like both?
4. What is x? I. x + y = 10. II. y = 4. (Answer A/B/C/D/E)
5. A round robin has 7 teams. How many matches?
6. How many arrangements does LEADER have?
7. Two cards are drawn without replacement. Find P(both aces).
8. A invests ₹5,000 for 12 months and B ₹6,000 for 10 months. Profit ₹9,000. A's share?
9. 'Sales rose after the ad, so the ad worked.' What weakens this?
10. All A are B; Some B are C. Does 'Some A are C' follow?
11. Find the sum of the first 15 terms of 2, 5, 8, ...
12. Find the remainder of 5^{100} divided by 6.
13. A can do a job in 20 days and B in 30 days. Together, how many days?
14. 60% like tea and 70% like coffee. At least what percent like both?

Answer key

1. Data Interpretation

1. 76
2. 72°
3. 25%
4. 66.67%
5. 2 : 3
6. 41.67%
7. −10%
8. 20%
9. 16%
10. Year 2 (25%)

2. Set Theory and Venn Diagrams

1. 30
2. 5
3. 10
4. 15
5. 30
6. 15
7. 55
8. 70%
9. 10
10. 65

3. Data Sufficiency

1. C
2. B
3. C
4. C
5. A
6. C
7. A
8. A
9. C
10. A

4. Analytical Puzzles

1. 15
2. Both are liars
3. Thursday
4. 5
5. Green
6. Apples only
7. Bina
8. 27
9. 15
10. 20

5. Advanced Permutation and Combination

1. 5,040
2. 60
3. 24
4. 115
5. 60
6. 35
7. 74
8. 15
9. 120
10. 165

6. Advanced Probability

1. 1/3
2. 1/6
3. 1/3
4. 15/16
5. 8/13
6. 5/42
7. 9/20
8. 3/4
9. 7/15
10. ₹3

7. Partnership

1. ₹3,000
2. 1 : 1
3. ₹4,500
4. ₹8,000

5. ₹3,000, ₹2,000, ₹1,000
6. 1 : 1
7. ₹6,000
8. ₹12,000
9. ₹2,000
10. ₹12,800

8. Critical Reasoning

1. Only I
2. Heavy rain
3. A third factor (e.g., discipline) causes both
4. A controlled test where plants without music grew slower
5. Dentists' recommendation is reliable
6. Water shortage is likely
7. Costs fell more or prices rose
8. Sita is not an A-grade student
9. Authorities should inspect and enforce emission norms
10. Crime also fell in nearby areas without CCTV

9. Advanced Syllogisms

1. Only I follows
2. Only II follows
3. Only I follows
4. Only I follows
5. Only I follows
6. Only II follows
7. Only I follows
8. I and II follow
9. Only I follows
10. Either I or II follows

10. Advanced Word Problems

1. 55
2. 81
3. 93
4. 7 h 20 min
5. 5 days
6. 1
7. 1 km/h
8. 13.33 L
9. 225
10. 70 s

Mini test answers

1. 33.33%
2. 54°
3. 8
4. C
5. 21
6. 360
7. 1/221
8. ₹4,500
9. Festival season began the same week
10. No
11. 345
12. 1
13. 12 days
14. 30%